

Lab 1 – Requirements Engineering

Problem Statement #47: Domain & SSL Certificate Expiry Alert System

Target Stakeholders / Actors: SysAdmin, Security Officer | Domain: Developer Tools & IT Operations

Requirements Table (FR-001–FR-005, NFR-001–NFR-002)

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall initiate TLS handshakes against monitored domains daily, extract certificate expiration dates, and trigger alerts at 30, 15, and 3 days before expiry.	High	Pass: Alert queued for a certificate expiring in 15 days. Fail: An expired certificate is ignored without notification.	Certificate expiry is the primary cause of unplanned outages and browser trust warnings; staged alerts give teams time to renew.
FR-002	Functional	The system shall perform a WHOIS lookup for each monitored domain daily, extract the domain registration expiry date, and flag any domain expiring within 30 days.	High	Pass: A domain expiring in 25 days appears in the daily flagged-domains report. Fail: A domain expiring in 10 days is absent from the report.	Unlike certificate expiry, a lapsed domain registration can take the entire site and associated services offline, so it must be tracked independently.
FR-003	Functional	The system shall escalate an unacknowledged alert from the assigned SysAdmin to the Security Officer if it remains unacknowledged 24 hours after being raised.	High	Pass: An alert open for 24h 1min is automatically routed to the Security Officer's escalation channel. Fail: An alert stays visible only to the original SysAdmin after 24 hours.	A fixed escalation ladder prevents a single missed notification from letting a domain or certificate expire silently.
FR-004	Functional	The system shall allow an authorized SysAdmin or Security Officer to acknowledge an open alert, recording the acknowledger's identity and a timestamp against that alert.	Medium	Pass: Acknowledging an alert sets its status to "Acknowledged" and stores user ID and timestamp. Fail: Alert status remains "Open" after the acknowledgment action is performed.	Provides an auditable trail of who is handling each alert and prevents unnecessary escalation of alerts already in progress.
FR-005	Functional	The system shall allow a SysAdmin to add, edit, or remove a domain or SSL endpoint from the monitored inventory through an administrative interface.	Medium	Pass: A newly added domain is included in the next scheduled audit run. Fail: A removed domain is still scanned during the next audit cycle.	Infrastructure changes constantly; the monitored list must stay accurate for the audits to be trustworthy.
NFR-001	Nonfunctional (Performance & Security)	The monitoring engine shall scan a list of 1,000 domains and SSL endpoints in under 3 minutes.	High	Pass: Benchmark tests confirm the target latency and security standards are met under simulated peak load.	Nightly audits must complete comfortably inside the maintenance window without overloading network or security infrastructure.
NFR-002	Nonfunctional (Reliability & Availability)	The monitoring engine shall successfully complete at least 99.9% of its scheduled daily audit runs (measured monthly), with automatic retry on transient network or WHOIS/TLS failures.	High	Pass: Monthly audit-run logs show $\geq 99.9\%$ of scheduled runs completed, with no two consecutive nightly runs missed. Fail: Two or more consecutive nightly audit runs fail to complete.	A missed nightly scan is equivalent to an undetected expiry, which directly defeats the system's core purpose.

Shaded rows = requirements supplied in the problem statement; remaining rows authored for this lab.